Lecture Transcript by Slide

Week 8 · Chapter 6: Junctional Rhythms · Atrial and Junctional Rhythms recording from 1:08:30 (42 min)
Auto-transcribed, so expect occasional errors, especially in technical terms. Timestamps are positions in the recording.

1:08:30 Chapter 6: Junctional Rhythms (title slide)

1:08:41 Electrical Flow in Junctional Arrhythmias

1:11:48 Junctional P Wave Preceding QRS

1:15:13 Electrical Flow in Junctional Arrhythmias / Normal Electrical Flow in Lead II

1:15:37 Mechanism of Premature Junctional Complex

1:17:05 Rules for Premature Junctional Complex

1:18:54 Premature Junctional Complex (ECG strip)

1:22:44 Mechanism of Junctional Escape Rhythm

1:24:23 Rules for Junctional Escape Rhythm

1:25:10 Junctional Escape Rhythm (ECG strip)

1:28:00 Mechanism of Accelerated Junctional Rhythm

1:29:52 Rules for Accelerated Junctional Rhythm

1:30:31 Accelerated Junctional Rhythm (ECG strip)

1:34:17 Mechanism of Junctional Tachycardia

1:35:06 Rules for Junctional Tachycardia

1:35:50 Junctional Tachycardia (ECG strip)

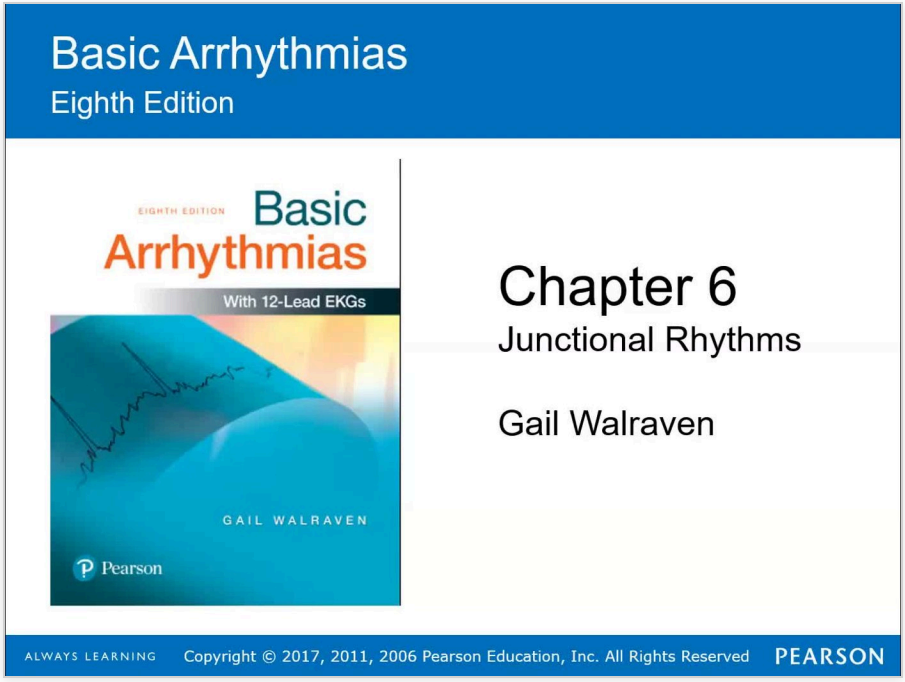
1:40:17 Supraventricular Tachycardia

1:41:11 Supraventricular Tachycardia: Overlapping Rate Ranges

1:43:14 Chapter 6 Practice Strips

1 Chapter 6: Junctional Rhythms (title slide)

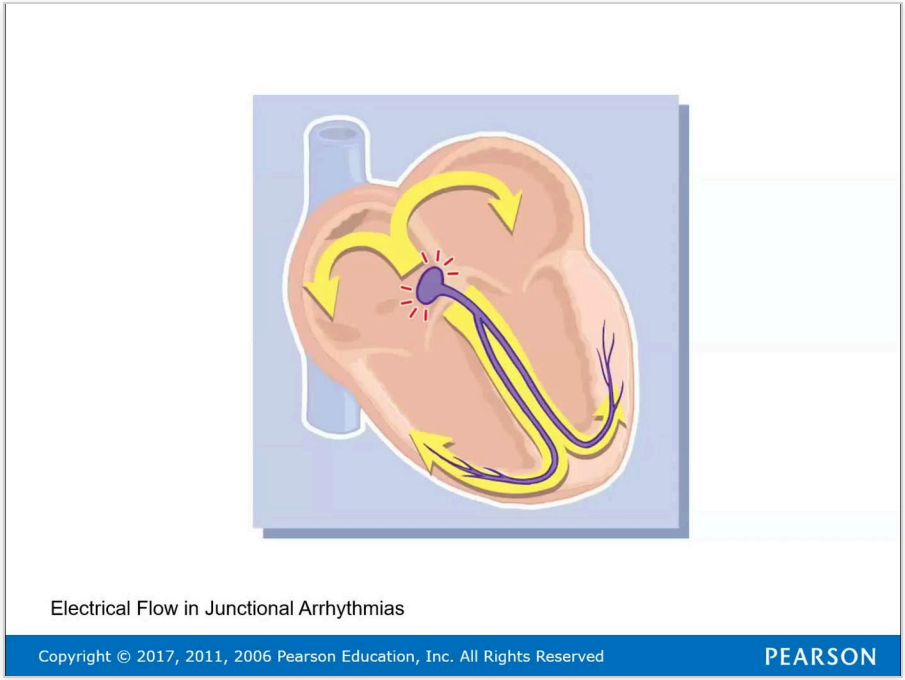
1:08:30 – 1:08:41



(Nothing was said while this slide was shown.)

2 Electrical Flow in Junctional Arrhythmias

1:08:41 – 1:11:48



1:08:42 Okay. So junctional rhythms come from the A-V node. Remember, A-V node, A-V junction, it means the same thing. Okay? So when the rhythm comes from the A-V node, it's going to send electrical signals in two different directions. One has to go backwards towards the right shoulder in order to depolarize the atria. The other one is going to go backwards. The other one is going to go down towards the left leg to depolarize the ventricles. So there's two key features here based on this physiology.

1:09:38 Number one, this is exactly why the P-waves are inverted. Because the right arm is a negative electrode. So in order to depolarize the atria, we have electrical flow going towards a negative electrode. And that's why the P-waves for junctional beats or junctional rhythms are inverted.

1:10:07 By the way, any time something has to go backwards in the body, we generally refer to that as retrograde. In this particular case, we call it retrograde conduction. Meaning the flow of electricity is going backwards, not in the normal direction that it flows. The other significant thing that happens physiologically here in this particular rhythm, beats or rhythms, is this is a timing thing.

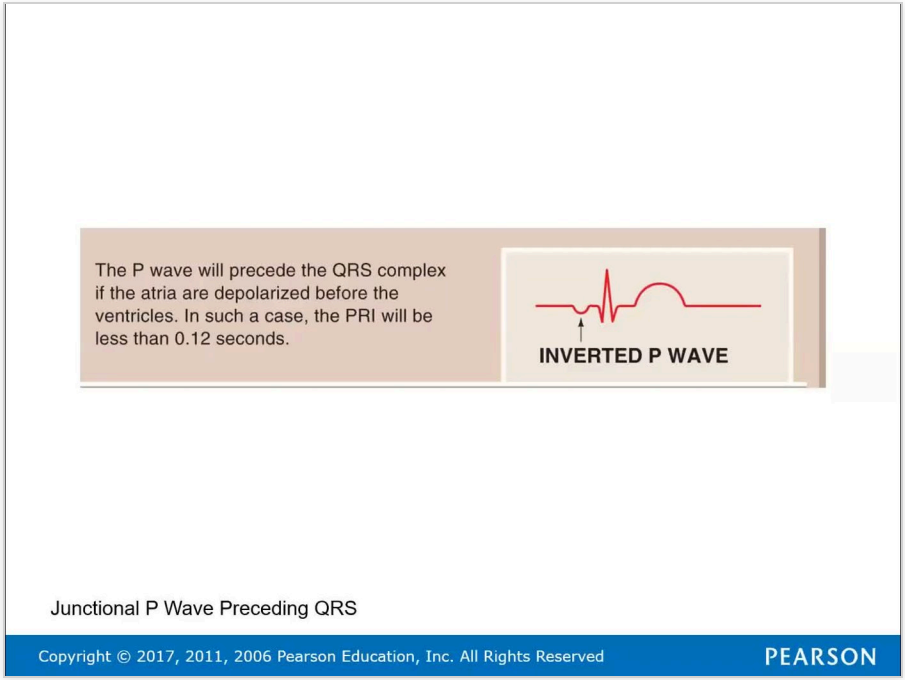
1:10:51 I got electrical flow going to the atria for atrial depolarization. I got electrical flow going down towards the ventricles for ventricular depolarization. Sometimes if the impulse reaches the atria, the atria will depolarize before the ventricles. Sometimes the atria and the ventricles depolarize all at the same time.

1:11:21 It depends on when the impulse reaches the atria and when the impulse reaches the ventricles. And then there's times where the ventricles will depolarize before the atria. So you can get any one of those three during a junctional beat or a junctional rhythm.

1:11:39 So that means it might displace the order of some of our waveforms. So I already talked about going towards a negative electrode.

3 Junctional P Wave Preceding QRS

1:11:48 – 1:15:13



1:11:53 So here is the timing of it, of it all. This inverted P wave will proceed or fall in front of the QRS if the atria are depolarized before the ventricles.

1:12:19 In this case, we'll have a short PRI, 0.12 seconds or less. But when the atria depolarized before the ventricles, we'll have an inverted P wave in front of the QRS.

1:12:38 If the atria and the ventricles are depolarized simultaneously, there will be no visible P wave. We will just see a narrow QRS and there will not be a P wave.

1:12:57 I am informing you right now, when you see a narrow QRS without a P wave, it can only be a junctional beat. So anytime you see a narrow QRS with no P wave, it can only come from the junction. Period. No exceptions.

1:13:27 So when they contract, when all four chambers contract or depolarize at the same time, the P wave is actually hidden inside the QRS and we do not see it.

1:13:44 If the ventricles depolarize before the atria, we will now see a QRS, a narrow QRS, followed by an inverted P wave after the QRS. Funky. Very funky. So you see this little indent down here? That's an inverted P wave.

1:14:11 So, junctional rhythms can have an inverted P wave before the QRS. Junctional beats can have no P wave because it's buried inside the QRS. Or we can have an inverted P wave after the QRS. The only time we can measure a PRI in these rhythms is when the P wave falls in front

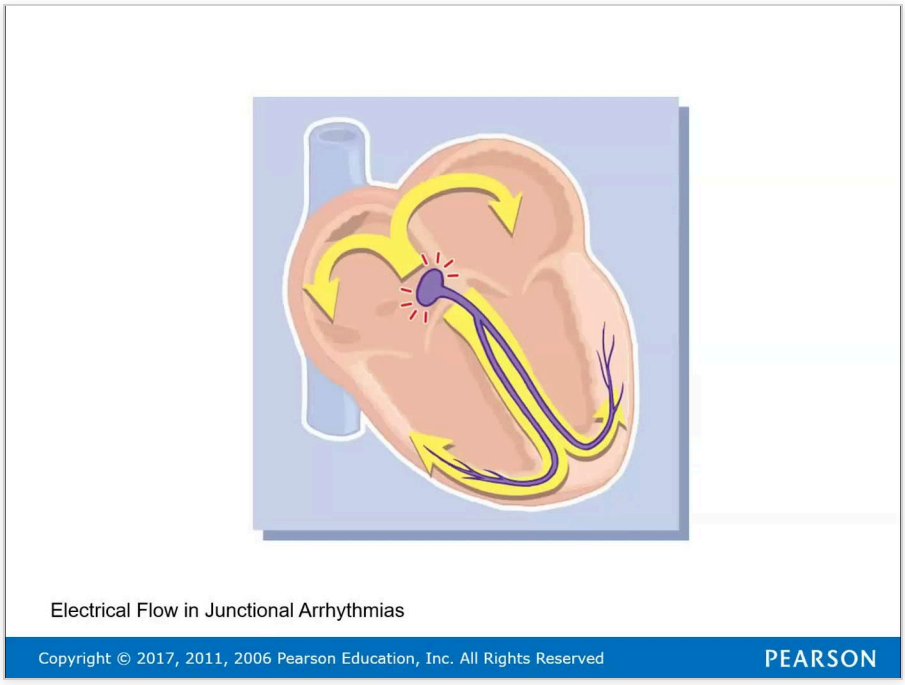
of the QRS. I can't measure a PRI if I can't see it.

1:14:41 And I can't measure a PRI if it comes after the QRS. Do you have ventricles depolarization at the same time as your papers depolarize? When they beat at the same time and they all contract at the same time. In which case you just see a QRS. No P wave.

1:15:06 I know it's confusing because it goes against the grain of what I've taught you so far. But because the beat is coming from the middle of the heart, it depends on what gets where first.

4 Electrical Flow in Junctional Arrhythmias / Normal Electrical Flow in Lead II

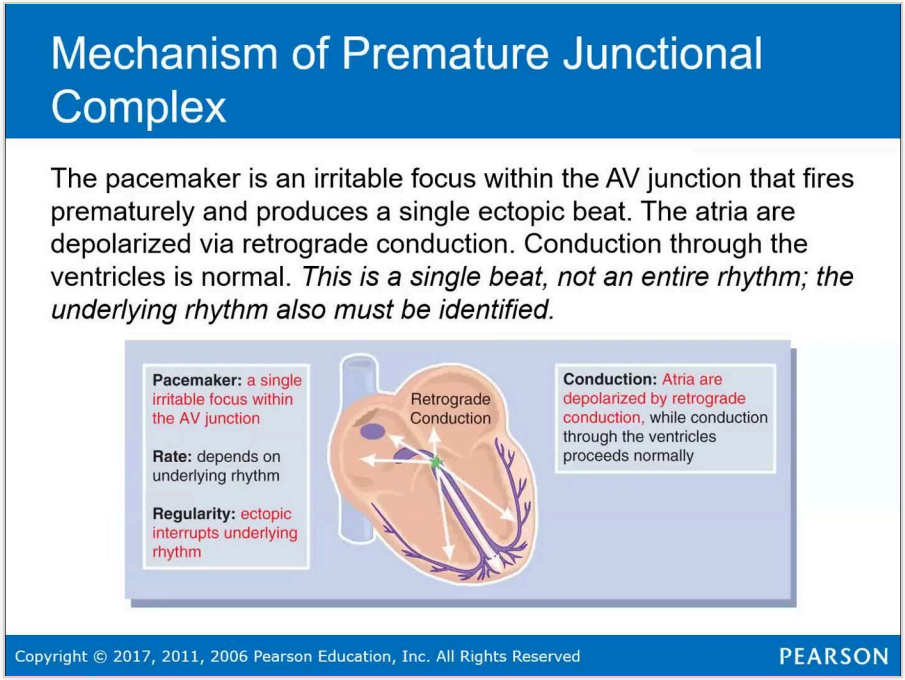
1:15:13 – 1:15:37



1:15:17 Right? You beat me in front of the line or I beat you in front of the line. Or we tie. It's really that simple. Right? Then you just have to understand where is it going to show up based on the timing of it all. Okay?

5 Mechanism of Premature Junctional Complex

1:15:37 – 1:17:05



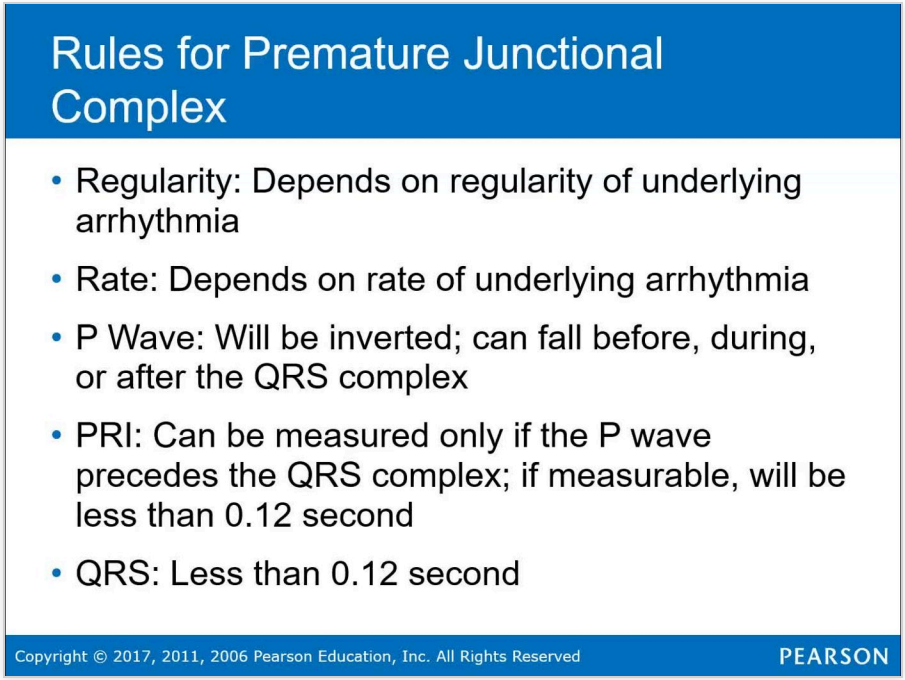
1:15:37 All right. Now we're going to talk about PJCs. Just like we talked about PACs, we're going to talk about PJCs. So, in the case of a PJC, this, again, is an irritable focal point. PJC, premature junctional contraction. So, that means an irritable focal point within the junction fires early or prematurely.

1:16:05 And it produces one single ectopic beat. In this particular case, though, the atria are depolarized via retrograde conduction because it's coming from the junction and has to go backwards to depolarize the atria.

1:16:27 And then it's still going to be a timing thing. I might see an inverted P wave before the QRS if the atria contract before the ventricles in this PJC. Or I might see a narrow QRS without a P wave because all four chambers depolarized at the same time. Or I might see a narrow QRS with a small P wave after the QRS for this PJC. This one single beat could show up any one of those three ways that I described. So, let's go through the rules.

6 Rules for Premature Junctional Complex

1:17:05 – 1:18:54



1:17:05 Regularity, again, because it's a single ectopic beat, it's going to depend on the underlying rhythm. And again, you know, to start out, I probably won't give you rhythms with more than one premature beat in it. But as we move on, I'm going to give you rhythms that have two PACs or two PJCs, right? The rules don't change. If you can find two R to R's with the same measurement, the underlying rhythm is regular. Then you just have to identify where are the ectopic beats coming from. And what's the key to identifying the ectopic beats, where they're coming from? P wave. Absolutely. If you see a positively deflected P wave on the ectopic beat, that has to come from the atria.

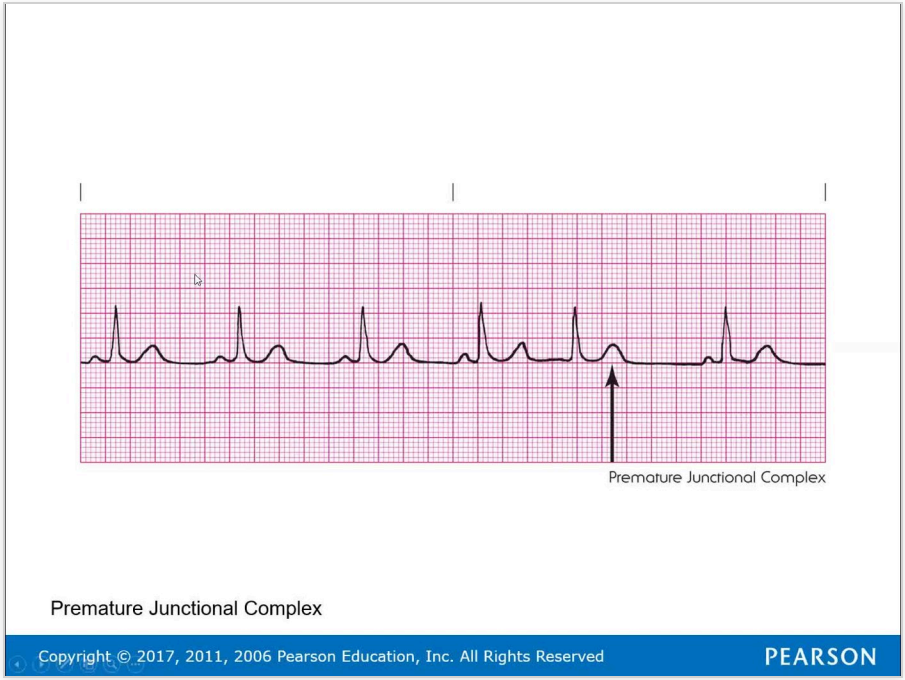
1:17:54 They're the only narrow complex beats that produce a positively deflected P wave. If it's coming from the junction, there's either A, no P wave, or an inverted P wave that falls in front or after the QRS.

1:18:12 Okay. So the regularity depends on the underlying rhythm. The rate is going to depend on the underlying rhythm. P waves. If we see them, we'll always be inverted. It can fall before the QRS. It can be buried within the QRS, which means I don't see it. Or it can fall after the QRS. I don't know how many times I can repeat myself.

1:18:34 I'm sounding like a broken record here. The PRI can only be measured if the inverted P wave falls in front of the QRS. And it will be short, 0.12 seconds or less. And the QRS will be normal or narrow.

7 Premature Junctional Complex (ECG strip)

1:18:54 – 1:22:44



1:18:54 Let's go through our information gathering. Just looking at this, you can see that the spacing, first two looks similar to the second two, looks similar to the third two. But then when I look at the fourth and fifth beat, that space seems to narrow. So that's an early beat. And then I have a compensatory pause. And then I get back to the underlying rhythm. So let's do our information gathering here. 5, 10, 15, 20, 21, 22, 23, 24, 25. 5, 10, 15, 20, 25. 5, 10, 15, 20, 23. And two halves make 24.

1:19:46 5, 10, 19, I'm sorry. 5, 10, 14, 18. And two halves make 19. 5, 10, 15, 20, 25, 30. 30 and a half. So regular, regular, regular, short, long. So regular, regular, regular, early beat, compensatory, pause, back to the underlying rhythm. Everybody follow me? Okay. So the underlying rhythm is regular with one ectopic.

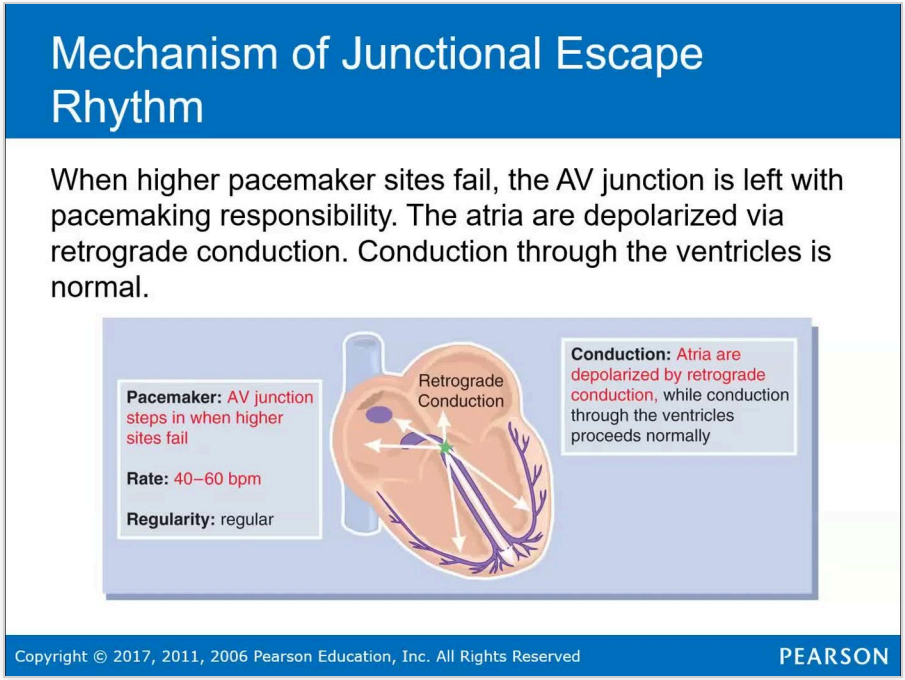
1:20:33 What did I say the spacing was? 25. So 1,500 divided by 25. Heart rate is 60. P waves. We look at these P waves. The regular beats all have these upright and rounded P waves that look the same. The ectopic beat has no P wave. So the underlying rhythm has P waves that are upright and uniform. This PRI is going to be 1, 2, 3, 4 and a half. 4 and a half. So that PRI is 4 and a half small boxes at 0.18.

1:21:22 This QRS is going to be 3, 1, 2, 3, 1, 2, 3. QRS is going to be 0.12. So the underlying rhythm is regular with a heart rate of 60, P waves that are upright and uniform, normal PRI, normal QRS. What is the underlying rhythm? No takers? Is everything normal? So this is normal sinus rhythm.

1:22:04 And then we have an early B. Where is this early B coming from? You see a narrow QRS. I don't see a P wave. So the P wave is going to be buried inside of this QRS, which means the atria and the ventricles depolarized at the same time. What did I tell you? A narrow QRS with no P wave can only come from where? The junction. So our interpretation here is normal sinus rhythm with 1PJC. All right.

8 Mechanism of Junctional Escape Rhythm

1:22:44 – 1:24:23

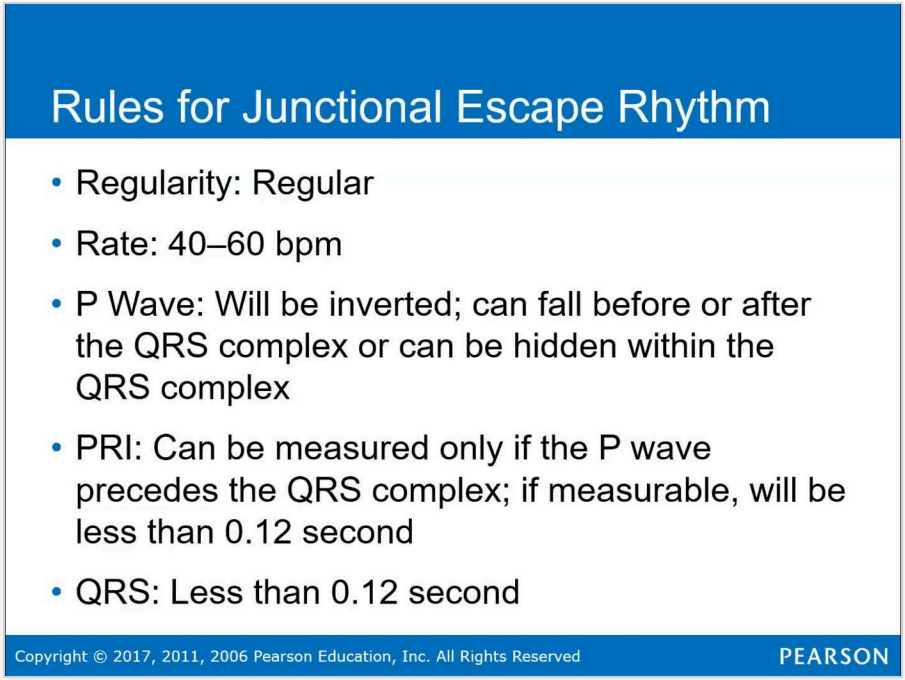


1:22:46 I'm going to keep moving. So the next rhythm we're going to talk about is something called J-E-R or junctional escape rhythm. In the beginning of the semester, I talked about early beats, which are premature beats, which are irritable beats. I also talked about late beats. Late beats occur when the pacemaker, the normal pacemaker, either fails or slows down. So something happens to the AV node, and it slows down. The first backup pacemaker to take over is the junction. So what junctional escape rhythm is, it's a backup pacemaker rhythm when the SA node either stopped firing or it fired at a rate lower than the junctional rate.

1:23:48 So when the higher pacemaker, the AV junction is now responsible to pick up the pacemaking responsibilities. because it's coming from the atria will be depolarized via retrograde conduction, which means it's going to be inverted P wave if we see it. But then the same timing rules apply. The P wave could be before, buried within, or after the QRS. And then the ventricles are depolarized normally, so it will be a narrow QRS.

9 Rules for Junctional Escape Rhythm

1:24:23 – 1:25:10

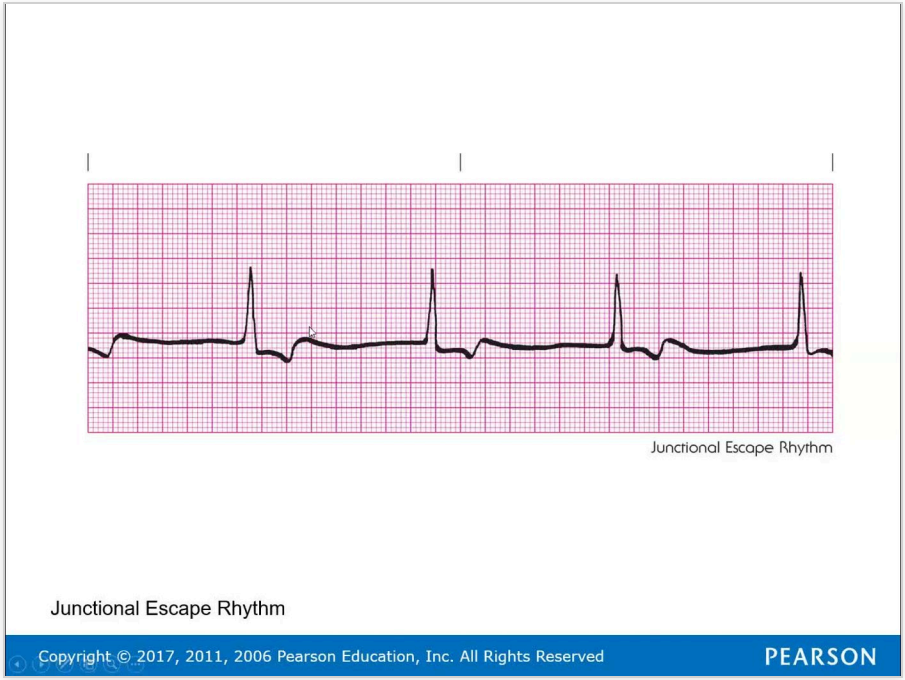


1:24:24 So this rhythm is always regular. The inherent heart rate for the junction is 40 to 60 beats per minute. You learned that already. We talked about that at the beginning of the semester.

1:24:39 The P waves will be inverted if we see them. The P wave can fall before the QRS if the atria depolarized before the ventricles. It can be buried within the QRS if they depolarize at the same time. Here I go as a broken record again. Or it can fall after the QRS if the ventricles depolarize before the atria. We can only measure the PRI if the inverted P wave falls in front of the QRS. And the QRS will be narrow. So let's take a look at this rhythm.

10 Junctional Escape Rhythm (ECG strip)

1:25:10 – 1:28:00



1:25:11 So let's go through our information gathering. 5, 10, 15, 20, 25, 30, 34, 35, 36. I would call that two halves and make it, I would say, 37 small boxes between the first and the second R wave. 5, 10, 15, 20, 25, 30, 35, 36. And two halves make 37. 5, 10, 15, 20, 25, 30.

1:25:48 33, 34, 35, 36. And two halves make 37. So this rhythm is regular. 1,500 divided by 37. Heart rate is 40. Does anybody see P waves? I don't see any P waves. So P waves, none. So if you see P waves, what should you, how should you describe the P waves if you see them in this rhythm?

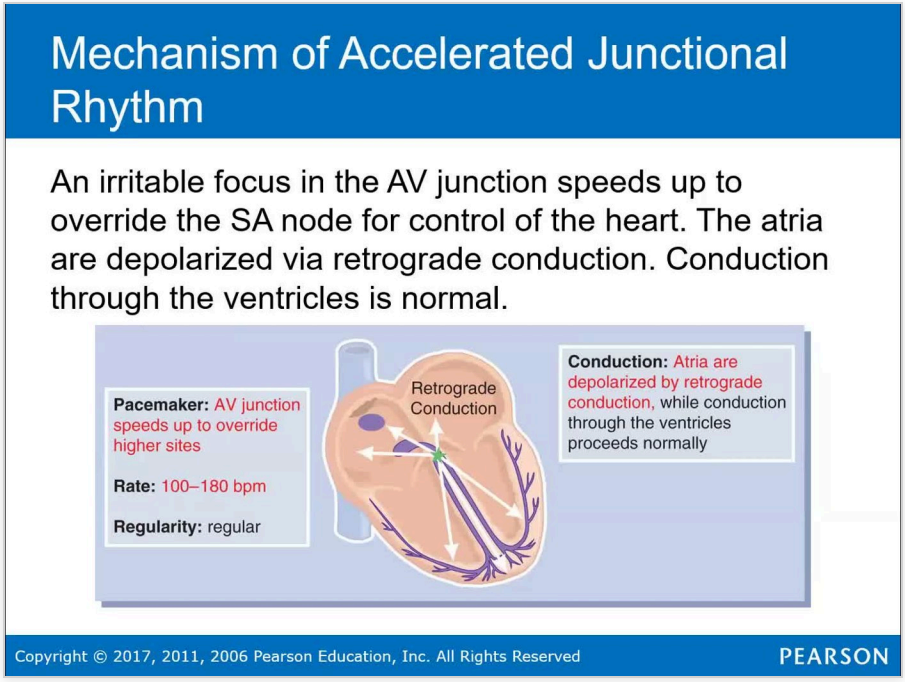
1:26:26 How did I tell you they appear? Inverted. So if you do see P waves, you're, and they're, and they're, you know, if you do see P waves, you're going to write inverted. If you don't see P waves, you're going to write none. So you think perceiving, or what would you just think? Well, it would be inverted if you see it before or after. You don't see it when it, when it occurs all at the same time. So we don't have to actually escape whether it's before or after.

1:26:55 Correct. I want to know that you know that I know that you know. I see you, see me, see you. Got it? Okay. So there's no P waves. We cannot measure PRI. This QRS looks like it's going to be three small boxes again. One, two, two and a half, maybe. One, two and a half. Yeah, it's two and a half. So 0.10. Now, I have a rhythm that's regular. I have a heart rate that's 40. I don't see P waves. And I have a narrow QRS. Narrow QRSs without a P wave can only come from where?

1:27:48 The junction, right? So our interpretation here is junctional escape rhythm or J-E-R. All right, next rhythm.

11 Mechanism of Accelerated Junctional Rhythm

1:28:00 – 1:29:52



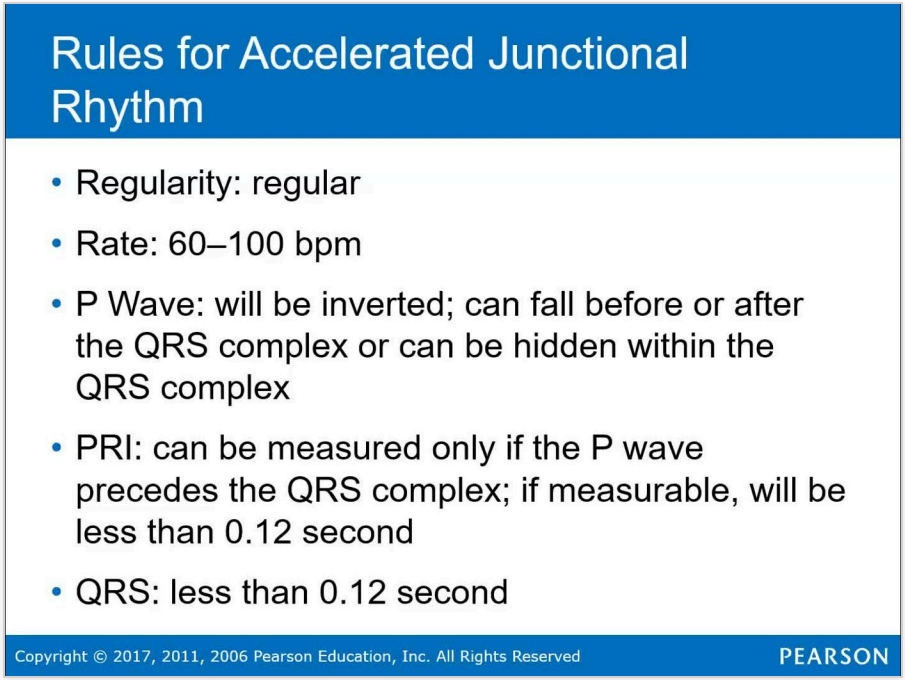
1:28:05 Now we're on to exceptions again. So the next rhythm we're going to talk about is accelerated junctional rhythm. So I know that I said to you guys, the inherent rate of the junction is 40 to 60. But the junction itself can become irritable. And when the junction becomes irritable, it can actually fire at a rate of 60 to 100.

1:28:38 And then the junction can sometimes become more irritable and fire at a rate of above 100. So in this case, AJR, accelerated junctional rhythm, it's an irritable focal point within the junction that fires repetitively at a rate faster than the SA node. So this is not an escape rhythm. This is not a backup pacemaker rhythm. This is an irritable rhythm. And it's irritable and it's firing at a rate faster than the SA node. So now it takes over as pacemaker. Again, fastest rate wins. Again, the HRD polarized via retrograde conduction. And conduction to the ventricles is normal. Now there's a typo on this slide. Please cross out the 100 to 180 beats per minute in red ink.

1:29:39 Well, it's red on the screen. It's not red on your PowerPoint. Please put in a rate of 60 to 100. Let's go through the rules.

12 Rules for Accelerated Junctional Rhythm

1:29:52 – 1:30:31

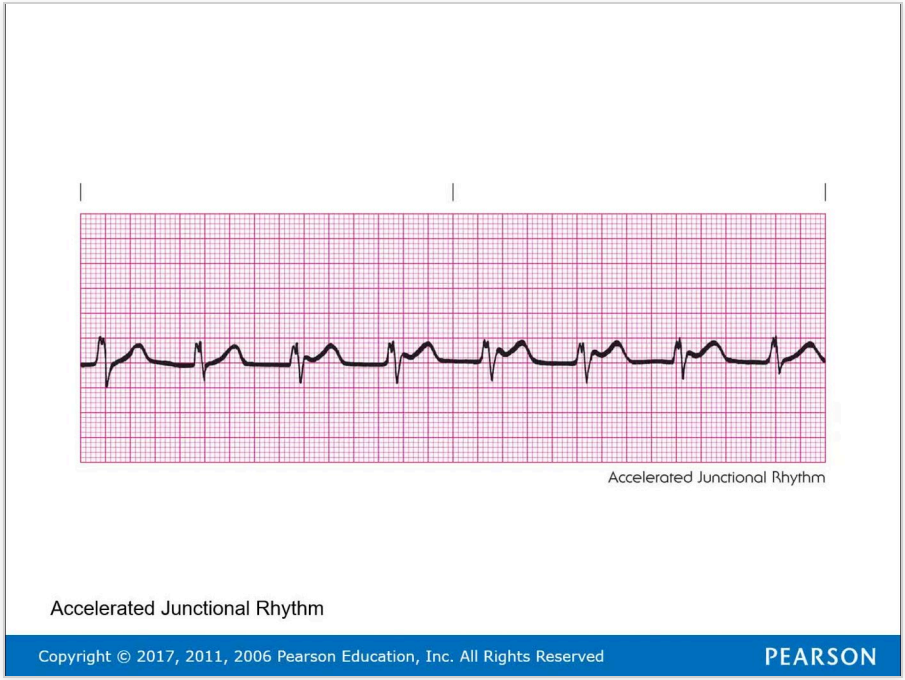


1:29:54 The junction always fires at regular intervals. So this rhythm will be regular. The heart rate for this accelerated junctional rhythm is 60 to 100.

1:30:06 P waves, if we see them, will be inverted. The P wave can fall before, buried within, or after the QRS, depending on the timing. We can only measure the PRI if the inverted P wave falls in front of the QRS. And the QRS will be normal or narrow. So let's look at this rhythm.

13 Accelerated Junctional Rhythm (ECG strip)

1:30:31 – 1:34:17



1:30:36 Now, this is kind of a funky looking QRS. Because it looks like it's got bunny ears at the top of the R wave. See the little hitch there? Okay.

1:30:54 So that's not something we're really going to discuss in class. That's really indicative of, like, a little slowdown in conduction in one of the bundle branches. So this is an intraventricular conduction delay, also known as an IVCD. Not something I'm expecting you to notice or be interpret. Those are difficult things to actually go through. So what we're going to do is we're going to count our R's from the same spot. So I'm going to go from the first point to the first point to the first point to the first point. It doesn't matter whether it's the first point or the second point, as long as you're going from the same point from one R to the next R to the next R.

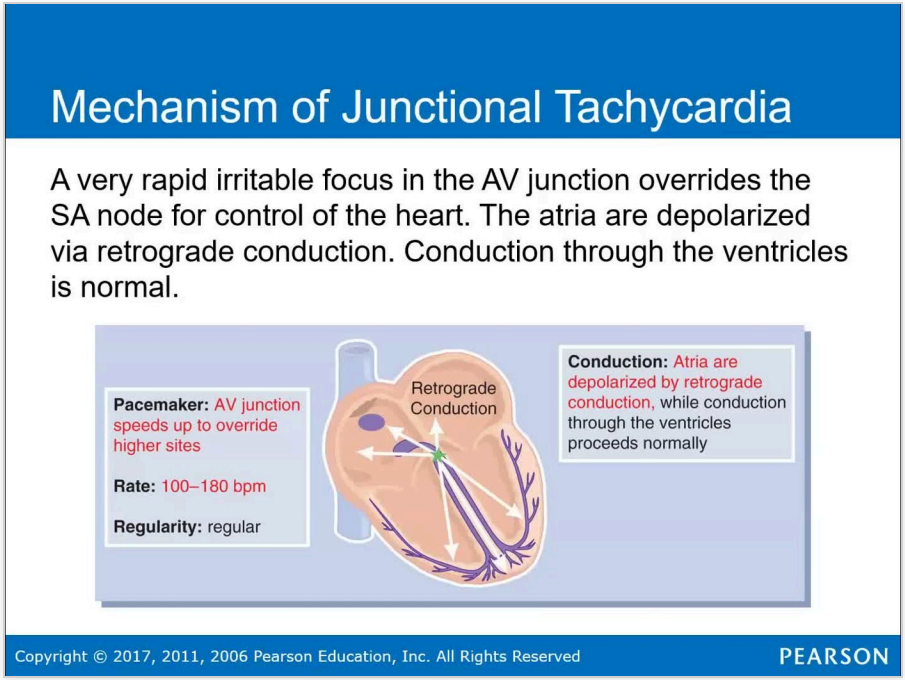
1:31:44 5, 10, 15, 16, 17, 18, 19 and a half for the second and third. 5, 10, 15, 16, 17, 18, 19 and two halves make 19. 5, 10, 15, 19 and a half. 5, 10, 15, 19 and a half. So this rhythm is regular. Was there more 19s or more 19s and a half? 19 and a half. So 1,500 divided by 19 and a half. Heart rate of 77. Anybody see any P waves? No P waves. No P waves, no PRI. Now we've got to measure our QRS. This is three small boxes. Well, the first one is three small boxes.

1:33:22 One, two, that's two and a half. One, two, three. One, two, three. It's three small boxes. So the QRS is 0.12. So we've got a narrow QRS with no P waves. It can only be coming from the junction. But in this case, the heart rate's not 40 to 60. It's 60 to 100. So our interpretation here

is accelerated junctional rhythm. Just be clear on the rate. I have a lot of students that kind of fumble on the rate as I teach this portion, right? If the heart rate's between 40 to 60, it's junctional escape. If the heart rate's between 60 to 100, it's accelerated junctional. All right.

14 Mechanism of Junctional Tachycardia

1:34:17 – 1:35:06

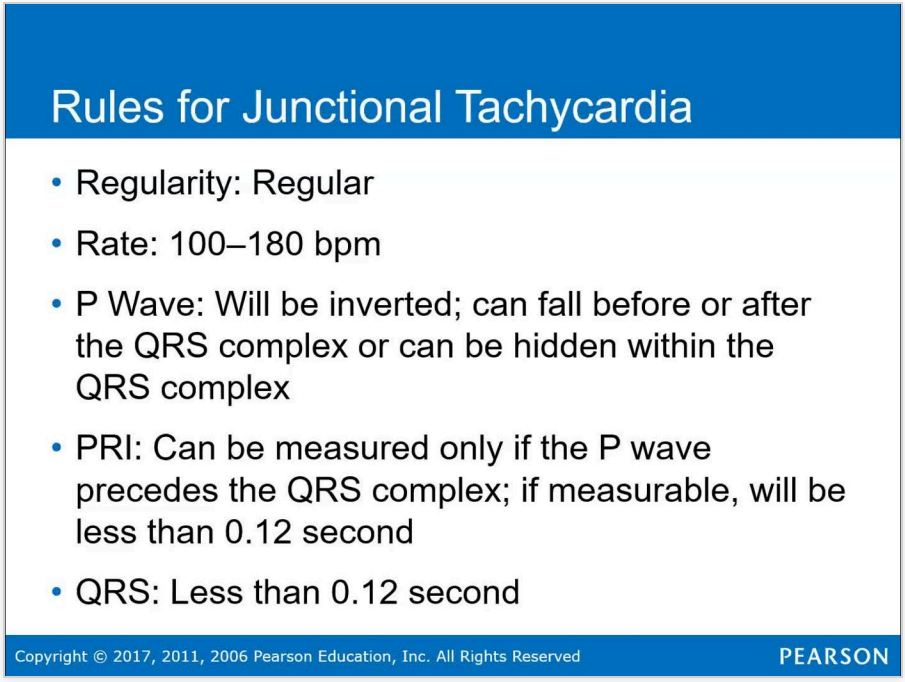


1:34:17 I'm going to move on to the next rhythm, which will be our last actual rhythm we'll talk about today. So this is now junctional tachycardia. So now the junction has become very irritable. And now it fires at a rapid repetitive rate of somewhere between 100 to 180 beats per minute. So when the AV node gets this fast, it's going to override the SA node as the pacemaker.

1:34:56 The atria will still be depolarized via retrograde, meaning they'll be inverted if we see them. And then the QRS will be normal or narrow.

15 Rules for Junctional Tachycardia

1:35:06 – 1:35:50

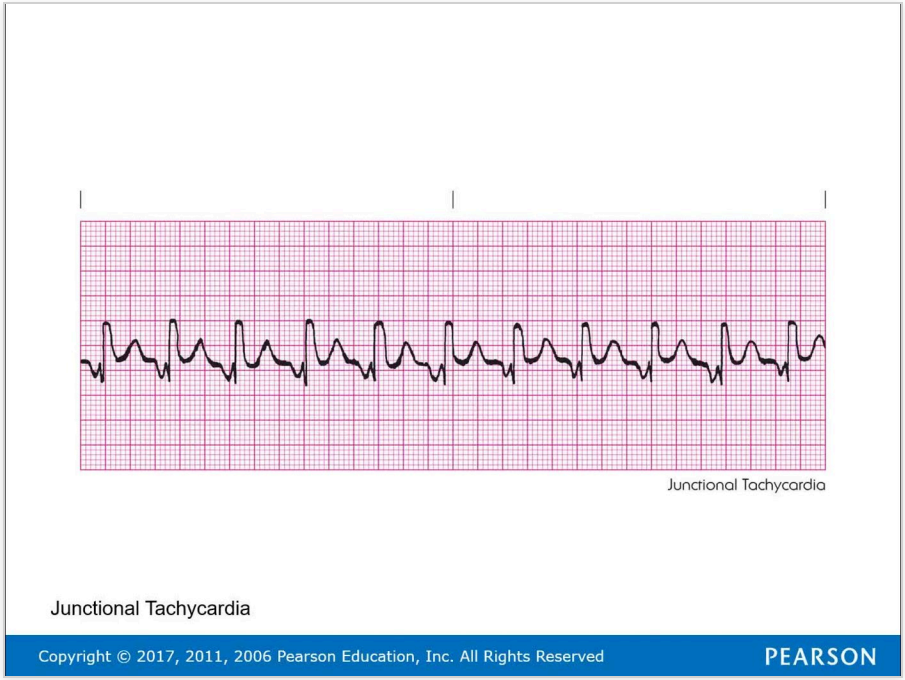


1:35:06 So let's go through our rules. Regularity here is going to be regular. The heart rate for this rhythm will be between 100 and 180. So there's actually going to be three separate rate ranges that you guys are going to need to know when it comes to junctional rhythms.

1:35:25 Junctional rhythms are always regular. Yes, always. P waves, if we see them, will be inverted, can fall before, be buried within, or after the QRS, depending on the timing. Measure the PRI only if the inverted P wave falls in front of the QRS, and the QRS will be normal or narrow.

16 Junctional Tachycardia (ECG strip)

1:35:50 – 1:40:17



1:35:51 And this is what it looks like. And this is what one version of it looks like, right? Remember, all of these junctional rhythms I can show you with a P wave or without a P wave. You have to be able to recognize that. So let's go through our information gathering here. So Do you see P waves in this rhythm? Yeah, these are inverted P waves right here. This is an inverted P wave before the QRS. Inverted P wave. And then this is my negative QRS. Inverted P, Q, R, S. Now, what's not necessarily part of the rhythm interpretation here is, again, this is more of a 12-lead interpretation thing, but there is an elevated ST segment in this rhythm.

1:36:57 Not that I'm expecting you to know that, but that you see the S wave, which ends right here, is supposed to be negatively deflected. The S wave never gets below the isoelectric line. This is the isoelectric line right there. So that S wave falls before it, and it goes right up into the T wave. So this is what's known as an elevated ST segment. This is the most common finding for somebody having an acute MI. ST elevation. All right.

1:37:37 So I'm just going to go from the middle of the R wave to the middle of the R wave. So 5, 10, 11, 12, 13 and a half. 5, 10, 11, 12, and two halves make 13. 5, 10, 11, 12, 13, 14.

1:38:03 5, 10, 14, 5, 10, 14. The next one is 14. The next one looks like it's 13 and a half. The next one is 14. That's 4, 9, 14 and a half. 13 and a half. So we go from 13 and a half to 14, or 14 and a half. So this rhythm is regular. I think 14 was the main number. So 1,500 divided by 14. 107.

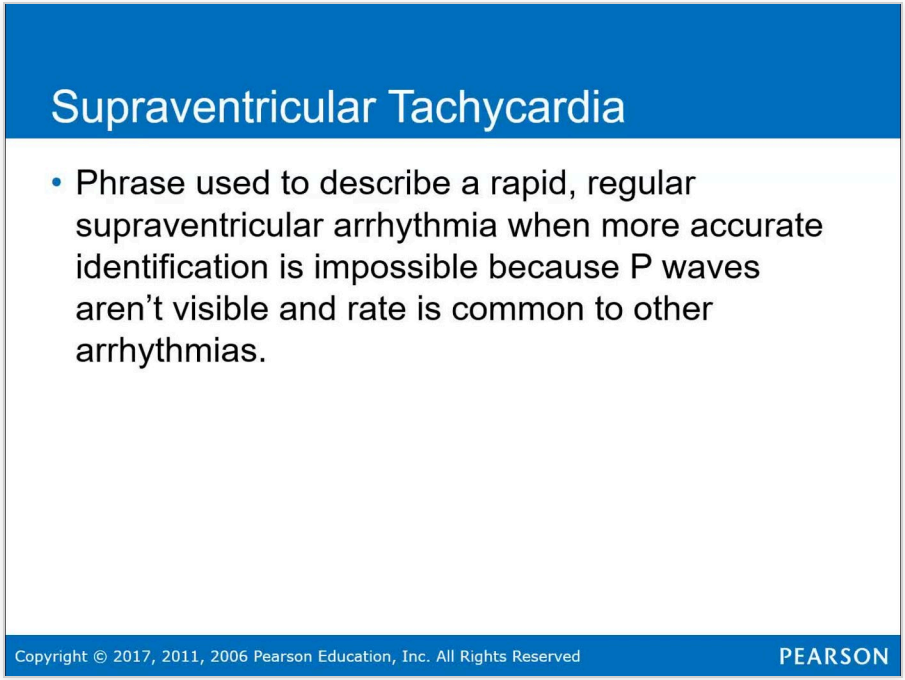
So the rhythm is regular. The heart rate is 107. P waves. Inverted. We've got to measure our PRI. So we're going to measure from the initial drop down of the P wave to the beginning of the Q. 1, 2 and a half.

1:39:17 1, 2, maybe 2 on that one. 1, 2 and a half. 1, 2 and a half. So 2 and a half. So this PRI is 0.10.

1:39:37 It's QRS is 1, 2, 3 small boxes. So I have a regular rhythm at a heart rate of 107. It's above 100. So automatically should be a red flag that says tachycardia. Above 100. Tachycardia. So I have a regular rhythm at 107. Inverted P waves. Can only be coming from junction, right? So I have a regular rhythm at 107 with inverted P waves. Automatically, this is junctional tachycardia. Got it? Okay. Got a couple more things that we're going to cover here.

17 Supraventricular Tachycardia

1:40:17 – 1:41:11

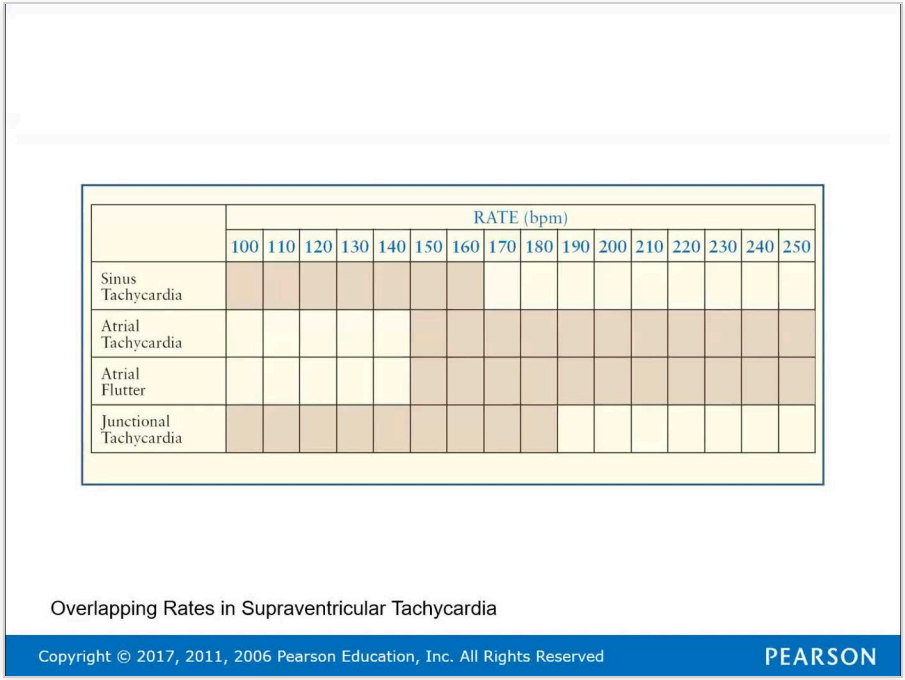


1:40:19 So I've already mentioned this term. Supraventricular tachycardia. Another way that I could say this is narrow complex tachycardia. Those beats that come from above the ventricle, the ventricles are depolarized via the normal conduction pathway, which means they will be narrow or normal.

1:40:47 So this is a phrase used to describe a rapid, regular. More P wave identification is impossible because they are not visible. The P waves are not visible. And the rate is in a common range to other tachycardias, narrow complex tachycardias. So the next slide will kind of help put this piece together.

18 Supraventricular Tachycardia: Overlapping Rate Ranges

1:41:11 – 1:43:14



1:41:15 And actually, I'm going to skip and jump to this slide because I like the highlighted pieces. So you can see four different rhythms listed on the slide.

1:41:29 From a heart rate range of 150 to 160, all four rhythms overlap that rate. So when I have a heart rate that's 155 with a narrow complex tachycardia, and I can't see the P wave, it could be any one of these four rhythms. It could be sinus tachycardia. It could be atrial tachycardia, atrial flutter, or it could be junctional tachycardia.

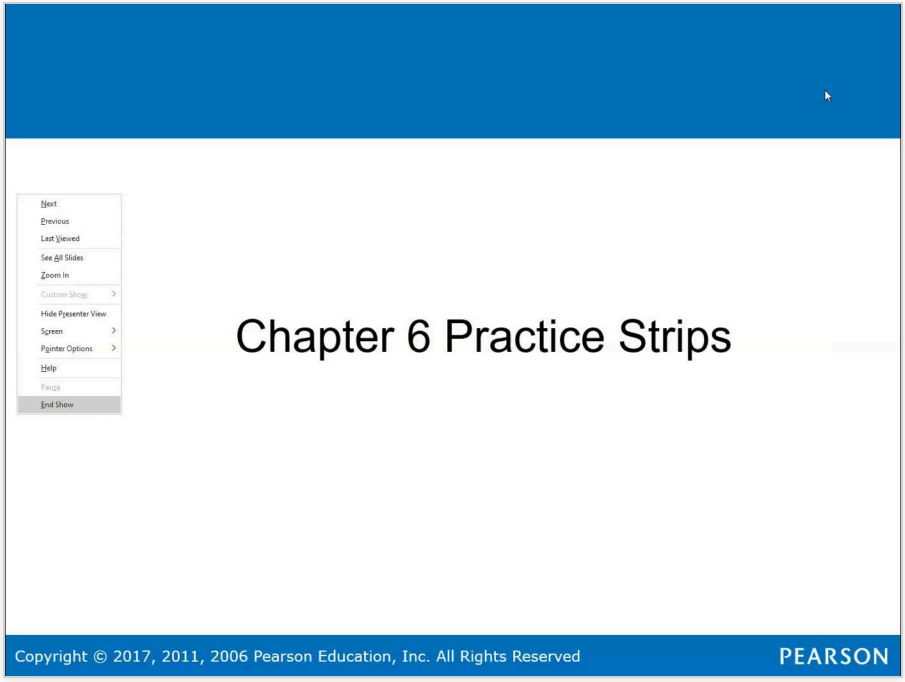
1:41:58 I don't know where it's coming from. So I can't label it as sinus tachycardia, or atrial tachycardia, or atrial flutter. So I have to interpret it as SVT, supraventricular tachycardia. So when you get rhythms that have narrow complex tachycardias, and you can't see the P wave, your interpretation should be SVT. And generally, that's going to be when the heart rate gets to 150 and above.

1:42:39 You can see that three of these rhythms overlap from 150 to 180. And then junctional tachycardia drops off at 180, and then we can have atrial tachycardia or atrial flutter that go all the way up to 250.

1:42:59 So we really don't know what rhythm it is. So when you get narrow complex with heart rates above 150 that you cannot see the P wave, I expect your interpretation should be SVT.

19 Chapter 6 Practice Strips

1:43:14 – 1:50:27



1:43:14 So if you want to see the P wave, I'm going to see the P wave, I'm going to see the P wave, I'm going to see the P wave, I'm going to see the P wave, I'm going to see the P wave. So if you want to see the P wave, I'm going to see the P wave. So if you want to see the P wave, I'm going to see the P wave, I'm going to see the P wave here. I think everybody got their homework except for you, right? Everybody saw your grade, your hourly grade posted in Brightspace and your homework grade posted in Brightspace.

1:43:55 I have everybody's homework that's due today. It is, where did I put you? You're here somewhere. There you go. Thank you. Um, I have a quick question. So I got everything finally, got my laptop and I got my ID and I couldn't click anything. But today I'm coming from a Myrtle, giant down Myrtle and you know Albany Man is on this side. Yes. Which way should I go to get to the West, West Lobby? So the West Lobby is on the other side. So the hospitals here, you're coming down Myrtle. So you would need to, are you, are you walking down or driving down? Driving.

1:44:41 So I'll probably park on Myrtle and then walk down. Is it a far walk from Myrtle or? Probably half a mile. It's not bad. It's really not bad. So you, if you park on Myrtle, it really depends on how far, where you are on Myrtle. Across from the doctor's office on Myrtle. Is that below South Lake or above South Lake? Above South Lake. So Myrtle and New Scotland, basically.

Like the entrance of the emergency, the adult emergency center, that side of Myrtle. Okay. Because I'm wondering if it's better to walk behind. At that point, you probably should, you probably should walk down Myrtle. Take a right on New Scotland.

1:45:41 Stay on the sidewalk. You're going to walk in front of the pillar entrance. You know, it has the big pillar entrance that they light up with colors at night. So park on Myrtle and walk on Myrtle until you get to New Scotland. Take a right on to New Scotland and walk all the way to the other side of the building. Okay. So on New Scotland, there's the Albany law school, both past the Albany law school. No, you're actually going to... So Myrtle is here, New Scotland is here, and New Scotland kind of bends just a little bit. Holland Avenue comes up this way, and the law school is over here.

1:46:26 Yeah. So as you come up and Holland comes up, you'll be able to walk on the sidewalk. Take a right as you get to the end of Albany Med's building. Just take a right, and you'll see where the entrance... It'll bring you right to the entrance. And is there like a number? Like... No, there's not a number. No, a number. Can you go to New Scotland? And this is... Because I work in the neurology department. So if you just put in the physician's facility, it'll take you right there. Physician's facility, right to the department. Okay. So it's not before the law school. It's like you turn right before the law school.

1:47:03 I think it's just before the law school. Yeah. Like essentially it's at the light, but... Yeah. Which actually the law school is kitty corner from where you want to go. Law school's here. Albany Med's corner of the building is right here. Now, when you go in that way, on the right hand side, you'll see an entrance on the right hand side. That's the one you want to go in. And then you'll see a sign that says Albany Medical College. So when you go right along the edge of the building, you're coming in. You'll see a wall that says Albany Medical College. You don't want to go in that entrance.

1:47:38 You want to go in the entrance on the right. The entrance before Albany Medical College. Correct. Once you get there, it'll be easy. Just walk to the other end of Albany Med's building and take a right. Okay. Okay. That's the West Lobby. Then I probably should park on New Scotland then because from Myrtle to that, that's a good part. Okay. You know. Yeah. Do we need to start doing some push-ups and sit-ups here? I mean, seriously, is it that far? Thank you. All right. I just needed my ID address for what happened. Yes, you do. Thank you for asking. You have a lab coat, right?

1:48:18 Yes. And then I just wear this on my coat. Did I give you the plastic sheet? No, no. Come with me. Okay. I'll get you what you need. I'll be right back in 30 seconds. Okay.

1:49:02 What can I do for you? I was wondering if I could look at my test, do that. I just wanted to know why I got wrong. What you would have to do is come before class. Okay. Like if you send me an email saying I'd like to look at my test before class in two weeks, I'd be happy to let you take a look at it. I don't hand them back because I want to make sure my tests are secure. Okay. But you're welcome to come and look at it and ask me questions. Okay. So if you send me an email and plan on getting here maybe at 1:30 in two weeks before class, then you could do that for a few minutes before class.

1:49:40 Okay. Sound good? Do I just go up to the office over there or? Yeah. So let me show you where my office is. Okay.

1:49:49 So what you can do is meet me right in the office at 1:30. Okay. So let me show you where my office is. Okay. So let me show you where my office is. Okay. So let me show you where my office is. Okay. So let me show you where my office is.